Use Case Flow

Student Record System

UC1 — Record and Update Attendance (FR-001)

Goal	Record new attendance or update previously recorded attendance for students in a selected subject and date.
Actor	Faculty
Main Flow (UC1)	
Steps	1. Faculty logs in to the system. 2. Faculty selects the option to manage attendance. 3. System prompts faculty to select a subject and date. 4. Faculty selects the subject and date. 5. System displays the students enrolled in the selected subject. 6. Faculty marks each student as Present or Absent, or selects an existing attendance record to update. 7. If updating, the system displays the previously recorded attendance. 8. Faculty enters or modifies the attendance details. 9. System validates the attendance entries. 10. System saves the attendance record or updates the existing record. 11. System confirms that the attendance has been saved successfully.
Alternate Flow (UC1)	
Incomplete Attendance	5a. System detects that attendance status is missing for one or more students. 5b. System displays an error message and asks faculty to complete the attendance. 5c. Return to step 6.
Exception Flow (UC1)	
Save Failure	10a. Attendance record fails to save. 10b. System displays an error message and asks faculty to retry. 10c. Return to step 8.

UC2 — Calculate Attendance Percentage (FR-002)

Goal	Calculate the attendance percentage for each student in a subject.
Actor	System
Main Flow (UC2)	
Steps	1. System retrieves the total number of classes conducted for the selected subject. 2. System retrieves the number of classes attended by the student. 3. System validates the attendance data. 4. System calculates the attendance percentage using the attendance data. 5. System displays the calculated attendance percentage.
Alternate Flow (UC2)	
No Attendance Data	3a. System detects that attendance data is unavailable. 3b. System displays a message indicating that no attendance data is available.
Exception Flow (UC2)	
Calculation Failure	4a. System fails to calculate the attendance percentage. 4b. System displays an error message. 4c. The calculation can be retried after correcting the problem.

UC3 — Enter and Update Marks (FR-003, FR-011)

Goal	Allow faculty to enter or update marks and ensure that entered marks are valid.
Actor	Faculty
Main Flow (UC3)	
Steps	1. Faculty logs in to the system. 2. Faculty selects the option to enter or update marks. 3. System prompts faculty to select a subject and student. 4. Faculty selects the subject and student. 5. System retrieves existing marks, if available. 6. Faculty enters new marks or updates the existing marks. 7. System validates the entered marks against the permitted minimum and maximum marks. 8. System saves the valid marks. 9. System confirms that the marks have been saved successfully.

Alternate Flow (UC3)	
Invalid Marks	7a. System detects marks below the permitted minimum or above the maximum permitted marks. 7b. System displays an appropriate error message. 7c. Faculty is asked to re-enter the marks. 7d. Return to step 6.
Exception Flow (UC3)	
Save Failure	8a. Marks fail to save. 8b. System displays an error message. 8c. Faculty can retry after the problem is resolved.

UC4 — Calculate Semester GPA (FR-004)

Goal	Calculate a student's semester GPA using marks, grade points, and subject credits.
Actor	System
Main Flow (UC4)	
Steps	1. System retrieves the student's marks for the selected semester. 2. System retrieves the corresponding subject credits. 3. System validates the marks and credit information. 4. System converts marks into grade points according to the institution's grading rules. 5. System calculates the semester GPA using grade points and subject credits. 6. System displays the calculated GPA for the selected semester.
Alternate Flow (UC4)	
Missing Marks or Credits	3a. System detects that marks or credits are missing for a subject. 3b. System displays a message indicating that the required information is unavailable. 3c. GPA calculation is postponed until the missing information is available.
Exception Flow (UC4)	
Invalid Academic Data	4a. System detects invalid marks, credits, or grading information. 4b. System displays an error message. 4c. The calculation can be retried after correcting the data.

UC5 — View Academic Records by Semester (FR-005)

Goal	Allow students to view their academic records and select a semester.
Actor	Student
Main Flow (UC5)	
Steps	1. Student logs in to the system. 2. Student selects the option to view academic records. 3. System displays the available semesters. 4. Student selects the required semester. 5. System retrieves the student's academic records for the selected semester. 6. System displays the student's attendance, marks, grades, and semester GPA. 7. System displays the student's CGPA.
Alternate Flow (UC5)	
No Records Available	5a. System detects that no academic records are available for the selected semester. 5b. System displays a message indicating that no records are available. 5c. Student may select another semester.
Exception Flow (UC5)	
Unauthorized Access	5d. System detects an attempt to access another student's academic records. 5e. System denies access and displays an appropriate error message.

UC6 — User Login and Logout (FR-006)

Goal	Allow users to access the system securely and end their sessions.
Actor	Student, Faculty
Main Flow (UC6)	
Steps	1. User opens the system. 2. System displays the login page. 3. User enters the required login credentials. 4. System validates the credentials. 5. System grants access to the appropriate user dashboard. 6. User performs the required activities. 7. User selects the logout option. 8. System terminates the user's session. 9. System returns to the login page.

Alternate Flow (UC6)	
Invalid Credentials	4a. System detects invalid login credentials. 4b. System displays an error message. 4c. User is asked to enter the credentials again.
Exception Flow (UC6)	
Login Failure	4d. System cannot verify the credentials because of a system error. 4e. System displays an appropriate error message and does not grant access.

UC7 — Search Students and View Profiles (FR-007)

Goal	Allow authorized faculty to search for students using their student IDs and view their profiles.
Actor	Faculty
Main Flow (UC7)	
Steps	1. Faculty logs in to the system. 2. Faculty selects the student search option. 3. System prompts faculty to enter a student ID. 4. Faculty enters the student ID. 5. System searches for the student. 6. System displays the matching student's profile details.
Alternate Flow (UC7)	
Student Not Found	5a. System cannot find a student matching the entered ID. 5b. System displays a message indicating that no matching student was found. 5c. Faculty may enter another student ID.
Exception Flow (UC7)	
Unauthorized Access	5d. System detects that the faculty member is not authorized to view the requested profile. 5e. System denies access and displays an appropriate error message.

UC8 — Maintain Subject Details and Credits (FR-008)

Goal	Maintain subject information and the corresponding credit values.
Actor	Authorized Faculty or Administrator
Main Flow (UC8)	
Steps	1. User logs in to the system. 2. User selects the subject management option. 3. System displays the available subjects. 4. User selects the option to add or update subject details. 5. User enters or modifies the subject name and credit value. 6. System validates the entered information. 7. System saves the subject details and credit value. 8. System confirms that the information has been saved successfully.
Alternate Flow (UC8)	
Invalid Credit Value	6a. System detects a missing, invalid, or non-permitted credit value. 6b. System displays an error message. 6c. User is asked to correct the information.
Exception Flow (UC8)	
Save Failure	7a. System fails to save the subject details. 7b. System displays an error message and allows the user to retry.

UC9 — Calculate Cumulative GPA (CGPA) (FR-009)

Goal	Calculate the student's CGPA using semester results and subject credits according to the institution's prescribed formula.
Actor	System
Main Flow (UC9)	
Steps	1. System retrieves the student's available semester results. 2. System retrieves the corresponding subject credits and grade-point information. 3. System validates the retrieved academic data. 4. System calculates the CGPA using the institution's prescribed formula. 5. System displays the calculated CGPA.

Alternate Flow (UC9)	
Incomplete Academic Records	3a. System detects missing semester results or required credit information. 3b. System displays a message indicating that the CGPA cannot be calculated with the available data. 3c. Calculation is performed when the required information becomes available.
Exception Flow (UC9)	
Calculation Failure	4a. System fails to calculate the CGPA. 4b. System displays an error message. 4c. The calculation can be retried after the problem is resolved.

UC10 — Filter Attendance Records (FR-010)

Goal	Allow faculty to filter attendance records by subject and date.
Actor	Faculty
Main Flow (UC10)	
Steps	1. Faculty logs in to the system. 2. Faculty selects the attendance records option. 3. System displays the available filtering options. 4. Faculty selects a subject and date. 5. System retrieves attendance records matching the selected filters. 6. System displays the filtered attendance records.
Alternate Flow (UC10)	
No Matching Records	5a. System finds no attendance records matching the selected filters. 5b. System displays a message indicating that no matching records were found. 5c. Faculty may change the filters and search again.
Exception Flow (UC10)	
Retrieval Failure	5d. System fails to retrieve attendance records. 5e. System displays an error message and allows faculty to retry.

UC11 — Generate Class-Wise Attendance Report (FR-012)

Goal	Generate a class-wise attendance report for faculty.
Actor	Faculty
Main Flow (UC11)	
Steps	1. Faculty logs in to the system. 2. Faculty selects the attendance report option. 3. System prompts faculty to select the required class or subject. 4. Faculty selects the class or subject. 5. System retrieves the corresponding attendance records. 6. System generates the class-wise attendance report. 7. System displays the generated report.
Alternate Flow (UC11)	
No Attendance Records	5a. System detects that attendance records are unavailable. 5b. System displays a message indicating that no report can be generated.
Exception Flow (UC11)	
Report Generation Failure	6a. System fails to generate the report. 6b. System displays an error message and allows faculty to retry.

UC12 — Notify Students of Low Attendance (FR-013)

Goal	Notify students whose attendance percentage falls below the institution's minimum attendance threshold.
Actor	System
Main Flow (UC12)	
Steps	1. System retrieves the student's attendance records. 2. System calculates the student's attendance percentage. 3. System retrieves the institution's configured minimum attendance threshold. 4. System compares the attendance percentage with the threshold. 5. System identifies students whose attendance is below the threshold. 6. System generates a low-attendance notification for the identified students. 7. System makes the notification available to the respective students.

Alternate Flow (UC12)	
Attendance Above Threshold	4a. System detects that the student's attendance meets or exceeds the minimum threshold. 4b. System does not generate a low-attendance notification for that student.
Exception Flow (UC12)	
Notification Failure	6a. System fails to generate or deliver the notification. 6b. System records or displays the failure so that the notification can be retried.

UC13 — View Subject-Wise Marks Summary (FR-014)

Goal	Allow faculty to view a summary of marks for students enrolled in a selected subject.
Actor	Faculty
Main Flow (UC13)	
Steps	1. Faculty logs in to the system. 2. Faculty selects the subject-wise marks summary option. 3. System displays the available subjects. 4. Faculty selects the required subject. 5. System retrieves marks for students enrolled in the selected subject. 6. System prepares the subject-wise marks summary. 7. System displays the summary to faculty.
Alternate Flow (UC13)	
Marks Not Available	5a. System detects that marks are not available for the selected subject. 5b. System displays a message indicating that no marks are available.
Exception Flow (UC13)	
Retrieval Failure	5c. System fails to retrieve the marks. 5d. System displays an error message and allows faculty to retry.

UC14 — Download Academic Records (FR-015)

Goal	Allow students to download their academic records as a report.
Actor	Student
Main Flow (UC14)	
Steps	1. Student logs in to the system. 2. Student selects the option to download academic records. 3. System retrieves the student's available academic records. 4. System prepares the academic report. 5. System provides the report for download. 6. Student downloads the report. 7. System confirms that the academic report is ready for download.
Alternate Flow (UC14)	
No Records Available	3a. System detects that no academic records are available. 3b. System displays a message indicating that the report cannot be generated.
Exception Flow (UC14)	
Report Generation or Download Failure	4a. System fails to generate the report or make it available for download. 4b. System displays an error message. 4c. Student may retry after the problem is resolved.